

密 专

浙江财经学院期中考试试卷

《数据库原理及应用（英）》课程试卷

考核方式： 闭 卷 考试日期： 20 年11月 日

适用专业、班级： 电子商务

题 号	一	二	三	四	五	六	七	八	九	十	总分
得 分											
评卷人											

(共 7 大题)

1. For each of the following questions there are four choices marked A, B, C and D. Only one of them is correct. You should decide on the correct choice and write down its mark on the answer sheet. (Points: 20)

- (1) Which of the following operations is supported by DDL ? (create, alter, drop)
DML(insert, update, delete, select),DCL(grant,revoke)
A . inserting some rows into a table B . modifying table structures in a database
C . finding some rows from a table D . deleting some rows from a table
- (2) In relational model, entity integrity means that _____.
A . there must not be empty rows in every relation
B . no attribute of a primary key can be null
C . attributes are allowed to be null
D . foreign keys are allowed to be null (三个完整性: entity integrity, referencial integrity

Business rule)

- (3) In SQL, in order to change the structure of a relation, we use statement.
A . UPDATE B . MODIFY C . ALTER TABLE D . CHANGE
- (4) Suppose we are given relation S(Sno CHAR(5), Sname CHAR(10), Sage INT), SC(Sno CHAR(5), Cno CHAR(5), Grade FLOAT) and C (Cno CHAR(5), Cname CHAR(40)), where a row in S means that a student with student number Sno, name Sname and age Sage, a row in SC means that a student with student number Sno enrolled in a course with course number Cno and obtains a grade Grade, and a row in C means that a course with course number Cno has course name Cname.

(4)-1 In order to list student name of each student that enrolled in the course 'ACCESS' and got a grade more than 90, the possible SQL

statement we may use will be .

A . SELECT Sname FROM (S join SC on S.Sno=SC.Sno) join C on SC.Cno=C.Cno where Cname= 'ACCESS' and Grade>=90

B . SELECT Sname FROM (S join C) join SC where Cname= 'ACCESS' and Grade>=90

C . SELECT Sname FROM (S join SC on S.Sno=SC.Sno) join C on SC.Cno=C.Cno where Grade>=90 AND Cname IN 'ACCESS'

D . SELECT Sname FROM (S join SC on S.Sno=SC.Sno) join C on SC.Cno=C.Cno where Cname= 'ACCESS'

(4)-2 In order to add column Sdept of S using SQL, we shall use command.

A . ADD Sdept TO S B . ALTER TABLE S ADD Sdept char(30)

C . UPDATE S Sdept D . ALTER TABLE S 'Sdept'

(4)-3 In order to create a view for the students that have never enrolled in any courses, we should use statement.

A . Create view S-no-enroll(Sno,Sname) as SELECT Sno, Sname FROM S WHERE Sno not in any courses

B . Create view S-no-enroll(Sno,Sname) as SELECT Sno, Sname FROM S WHERE Sno not in(select Sno from SC)

C . Create view S-no-enroll(Sno,Sname) as SELECT Sno,Sname FROM S (left) JOIN Sc on S.Sno=Sc.Sno WHERE Cno IS NULL

D . Create view S-no-enroll(Sno,Sname) as SELECT Sno,Sname FROM S WHERE Cno IS NULL

(5) Which of the following operations can retrieve from a relation all records that satisfy a condition to form a new relation?

A . Projection B . Join C . Selection D . Division

(6) In SQL, in order to add a new column CN into base table S, we use statement.

A . ADD TABLE S(CN CHAR(8)) B . ADD TABLE S ALTER(CN CHAR(8))

C . ALTER TABLE S ADD CN CHAR(8) D . ALTER TABLE S (ADD CN CHAR(8))

(7) A data model comprises three components, they are .

A . the external schema, the conceptual schema and the internal schema(三层结构)

B . the relations, the tuples and the primary keys (关系模型中的几个重要概念)

C . the structural part, the manipulative part and the integrity rules

D . the meaning of model, the operational rules and the implementation language

(8) In SQL, views are corresponding to the ____ schema of a database.

A . external B . conceptual C . inner D . storage

2. Calculation of relational algebra (2.5points each, total 20points).

Suppose we are given relation *R* and *S* as below, please give results of the following relational algebra expressions.

$R(X,Y) \div s(Y,Z) \quad P(X)$

$T(AC,B) \div U(B,D) = \{(1,3), (2,4)\}$

T			U		V	
A	B	C	B	D	C	E
1	2	3	1	3	3	1
1	1	3	2	4	3	2
2	1	4	1	5	4	6
2	2	4	2	6	4	7
3	4	5	1	9		
3	1	4	2	7		

1) $\tilde{O}_B(T) \cap \tilde{O}_B(U)$

B
1
2

2) $\tilde{O}_B(T) \cup \tilde{O}_B(U)$

B
1
2
4

3) $\tilde{O}_B(T) - \tilde{O}_B(U)$

B
4

4) $T \bowtie V$

A	B	C	E
1	2	3	1
1	2	3	2
1	1	3	1
1	1	3	2

2	1	4	6
2	1	4	7
2	2	4	6
2	2	4	7
3	1	4	6
3	1	4	7

5) $\sigma_{T.B=U.B=2}(T \times U)$

A	T.B	C	U.B	D
1	2	3	2	4
1	2	3	2	6
1	2	3	2	7
2	2	4	2	4
2	2	4	2	6
2	2	4	2	7

8) $T \div U = \{(1,3),(2,4)\}$

A	C
1	3
2	4

3. (60%) The following tables form part of a database held in a relational DBMS.

Product(**P_number**, Name, type, cost, price, description, produce_time, useful-life)

Customer(**Cust_Number**, Fname, Lname, Address, profession, birthday, sex)

Order(**Ord_number**, **Cust_Number**, date, Date_due, Total_price, Ord-status)

Details(**Ord_number**, **P_number**, quantity, status)

Where **Product** contains product details and P_number is the primary key. **Customer** contains customer details for each hotel. Each customer can play one or more orders and each order is played by one customer. Each **order** includes one or more **products**. The statue of order denotes whether the order is completed, valued as ‘unaffirmed’(未确认), ‘affirmed’ (已确认), ‘outstanding’ (已付款待送货), ‘delivered’ (已送货) and ‘invoice’ (已开发票) .

(a) identify the FKs

Order : foreign key(Cust_Number) references Customer(Cust_Number)

Details:foreign key(Ord_number) references Order(Ord_number)

Foreign key(P_number) references Product(P_number)

Π order_number,date,Date-due,Fname, Lname,address,P_number, Name, quantity,
status (δ Ord-status=' outstanding'(Customer $\bowtie$ Order $\bowtie$ Details $\bowtie$ Product))

(a) Formulate the following queries in RA and SQL respectively: (point:30)

(1) List full details of all customers in 'shanghai'.

δ address like 'shanghai'(Customer)

Select *

From Customer

Where address like 'shanghai'

(2) List all the orders that are 'outstanding', include order_number, date, Date -due, customer's name, address, product and status, order by date.

Π order_number,date,Date-due,Fname, Lname,address,P_number, Name, quantity, status
(Customer $\bowtie$ δ Ord-status = ' outstanding'(Order) $\bowtie$ Details $\bowtie$ Product)

Select order_number,date,Date-due,Fname, Lname,address,P_number, Name, quantity, status

From((Customer Cjoin Order OR on C. Cust_Number=OR. Cust_Number)

Join Details on OR. Ord_number=Details. Ord_number)

join Product Details. P_number=Product. P_number

Where Ord-status = ' outstanding'

(3) List the information of customers who have never played an order.

Π cust_number (Customer)- Π cust_number (Order)

Select *

From Customer

Where cust_number not in(select cust_number
From Order)

Select *

From Customer

Where not exists(

Select *

From Order

Where Order.cust_number=Customer.cust_number)

- (4) List the numbers, names and quantities of products that bought by the customer named 'John'.

$\pi_{P_number, Name, quantity} (\delta_{Fname='John'}(Customer) \bowtie Order \bowtie Details$
 $\bowtie Product)$

Select P.P_number, Name, quantity

From ((Customer C join Order O On C.custnumber=O. custnumber) JOIN

Details D ON O.Ord_number=D. Ord_number) join Product P on D.P_number=P.
P_number

Where Fname='John'

- (5) List the numbers,names of products that have been bought by all users

$\pi_{P_number, cust_number}(Order \bowtie Details) \div \pi_{cust_number} (customer)$

LIST the numbers,names of customers that have bought all fresh milks

$\pi_{cust_number,P_number} (Order \bowtie Details) \div \pi_{P_number} (\delta_{type='fresh\ milk'}($
product))

逐个判断每个产品，不存在任何一个顾客，不存在所指定产品指定客户的订单记录

Select *

From Product

Where not exist(

Select *

From customer C

Where not exists(

Select *

From Order join Details on Order.Ordnumber=Details.Ordnumber

Where Order.custnumber=C.custnumber

And Details.P_number=Product.P_number

)

)

(b) Formulate the following questions in SQL: (point:30)

(6) How many products of each type?

Select **type**, count(P_number)

From Product

Group by **type**

(7) What is the total revenue today?

Select sum(Total_price)

From Order

Where date=getdate()

(8) How many kind of products that have never purchased by customers?

Select count(P_number)

From Product

Where P_number not in(

Select P_number

From Details)

Select count(*)

From Product

Where P_number not in(

Select P_number

From Details)

(9) Move all the orders and their details with a date before 2010-12-31 from Order and Details to their archive tables called Order_archive and Detail_archive, the archive table have same structure.(First insert the information into two new tables ,and then delete from) (12points)

Insert into Order_archive

Select *

From Order

Where date<'2010-12-31'

Insert into Detail_archive

Select *

From Detail

Where Ord_number in

(select Ord_number

From Order
Where date<'2010-12-31')

Delete from Detail
Where Ord_number in
(select Ord_number
From Order
Where date<'2010-12-31')

Delete from Order
Where date<'2010-12-31'

- (10) Create a view of all information of products, including product names, types, prices, colors.

Create view P-view(names, types, prices, colors)

as

Select names, types, prices, colors

From Product

- (11) List the customers who have played orders more than 20 times by numbers and names.

Select *

From Customer

Where cust_number in

(Select cust_number

From Order

Group by cust_number

Having count(Ord_number)>20

)

Select cust_number

From Order

Group by cust_number

Having count(*)>20

Select cust_number,custName,Address

From Customer C join Order ON C.custnumber=Order.custnumber

Group by cust_number,custName,Address

Having count(Ord_number)>20